

DEPARTMENT OF INFORMATION TECHNOLOGY

24ITT32 - PROGRAMMING IN JAVA

INDIVIDUAL PROJECT ON
ACADEMIC MANAGEMENT SYSTEM

P R O J E C T R E P O R T

Submitted by

SURIYAPRASATH K
25ITR148

JAVA ASSIGNMENT

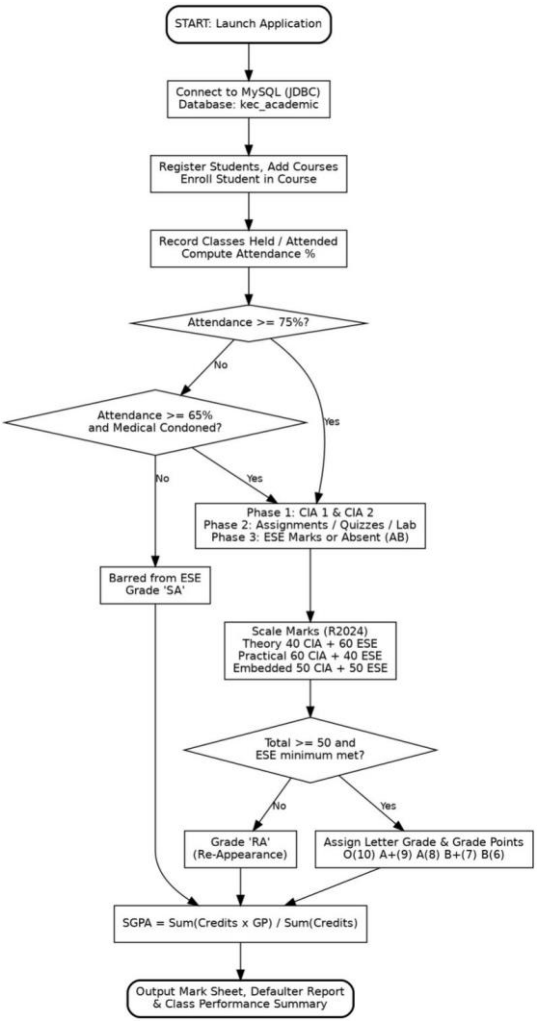
AIM:

Development of Kongu Engineering College (R2024) Student Academic & Continuous Evaluation System using Java JDBC.

OBJECTIVE:

Design and develop a database-driven Java application using JDBC to manage student records, perform incremental marks entry (Continuous Internal Assessments and End Semester Examinations), enforce attendance eligibility rules, and generate formal academic performance reports according to Kongu Engineering College (Autonomous) - Regulation 2024.

FLOW CHART:



PROCEDURE:

- Load the MySQL Connector/J driver so the Java application can communicate with the MySQL database.
- Establish a connection to the kec_academic database on the local MySQL server using DriverManager.getConnection().
- Store and retrieve student details (roll number, name, department, semester) and course details (course code, title, credits, course type).
- Enroll students in semester courses, record the number of classes held and attended, and calculate the attendance percentage.
- Verify examination eligibility from attendance: 75% or above is eligible, 65%-74.9% requires medical condonation, and below 65% is marked Shortage of Attendance (SA) and barred from the End Semester Exam.
- Record CIA Test 1, CIA Test 2 and assignment / quiz / lab marks, and scale the internal marks according to the course type weightage (Theory 40:60, Practical 60:40, Embedded 50:50).
- Record End Semester Exam marks out of 100 (or mark the student Absent), check the minimum ESE requirement, and scale the marks according to the course type.
- Combine scaled CIA and ESE marks to get the total, assign the letter grade and grade points, and award RA (Re-Appearance) if the student fails or scores below the pass mark.
- Generate the student semester mark sheet (SGPA and equivalent percentage), the attendance defaulter report, and the class performance and result summary on the command prompt console.

DATABASE SCHEMA:

1. STUDENTS TABLE:

```
mysql> DESCRIBE students;
+------------------+--------------+------+-----+---------+-------+
| Field            | Type         | Null | Key | Default | Extra |
+------------------+--------------+------+-----+---------+-------+
| roll_no          | varchar(15)  | NO   | PRI | NULL    |       |
| name             | varchar(100) | NO   |     | NULL    |       |
| department       | varchar(20)  | NO   |     | NULL    |       |
| current_semester | int          | NO   |     | NULL    |       |
+------------------+--------------+------+-----+---------+-------+
4 rows in set (0.312 sec)
```

2. COURSES TABLE:

```
mysql> DESCRIBE courses;
+-------------+-----------------------------------------+------+-----+---------+-------+
| Field       | Type                                    | Null | Key | Default | Extra |
+-------------+-----------------------------------------+------+-----+---------+-------+
| course_code | varchar(15)                             | NO   | PRI | NULL    |       |
| title       | varchar(100)                            | NO   |     | NULL    |       |
| credits     | int                                     | NO   |     | NULL    |       |
| course_type | enum('THEORY','PRACTICAL','EMBEDDED') | NO   |     | NULL    |       |
+-------------+-----------------------------------------+------+-----+---------+-------+
4 rows in set (0.041 sec)
```

3. ENROLLMENTS TABLE:

```
mysql> DESCRIBE enrollments;
+------------------+-------------+------+-----+---------+----------------+
| Field            | Type        | Null | Key | Default | Extra          |
+------------------+-------------+------+-----+---------+----------------+
| enrollment_id    | int         | NO   | PRI | NULL    | auto_increment |
| roll_no          | varchar(15) | NO   | MUL | NULL    |                |
| course_code      | varchar(15) | NO   | MUL | NULL    |                |
| semester         | int         | NO   |     | NULL    |                |
| classes_held     | int         | NO   |     | 0       |                |
| classes_attended | int         | NO   |     | 0       |                |
| medical_approved | tinyint(1)  | NO   |     | 0       |                |
+------------------+-------------+------+-----+---------+----------------+
7 rows in set (0.040 sec)
```

4. ASSESSMENT MARKS TABLE:

```
mysql> DESCRIBE assessment_marks;
+---------------+--------------+------+-----+---------+-------+
| Field         | Type         | Null | Key | Default | Extra |
+---------------+--------------+------+-----+---------+-------+
| enrollment_id | int          | NO   | PRI | NULL    |       |
| cia1          | decimal(5,2) | YES  |     | NULL    |       |
| cia2          | decimal(5,2) | YES  |     | NULL    |       |
| activities    | decimal(5,2) | YES  |     | NULL    |       |
| ese           | decimal(5,2) | YES  |     | NULL    |       |
| ese_absent    | tinyint(1)   | NO   |     | 0       |       |
+---------------+--------------+------+-----+---------+-------+
6 rows in set (0.048 sec)
```

SOURCE CODE:

MAINAPP.JAVA:

```
import java.sql.Connection;
import java.util.Scanner;
/**
 * MainApp - Entry Point for the Kongu Engineering College (Autonomous) - R2024
 * Student Academic & Continuous Evaluation System using Java JDBC.
 */
public class MainApp {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        StudentDAO studentDAO = new StudentDAO();
        AcademicDAO academicDAO = new AcademicDAO();
        printBanner();
        // Initial Connectivity Check
        try (Connection con = DBConnection.getConnection()) {
            System.out.println("[Database] Successfully connected to KEC Academic Database
                (kec_academic).");
        } catch (Exception e) {
            System.err.println("[Database Warning] Could not connect automatically: " + e.getMessage());
            System.err.println("Ensure MySQL Server is running on port 3307 or 3306.");
        }
        boolean running = true;
        while (running) {
            printMenu();
            System.out.print("Enter your choice [0 - 14]: ");
            String input = sc.nextLine().trim();
            switch (input) {
                // Section 1: Student & Course Management
                case "1":
                    studentDAO.addStudent(sc);
                    break;
                case "2":
                    studentDAO.listStudents();
                    break;
                case "3":
                    studentDAO.addCourse(sc);
                    break;
                case "4":
                    studentDAO.listCourses();
                    break;
                case "5":
                    studentDAO.enrollStudent(sc);
                    break;
                case "6":
                    studentDAO.listEnrollments();
                    break;
                case "7":
```

```
                studentDAO.recordAttendance(sc);
                break;
            case "8":
                studentDAO.approveMedicalCondonation(sc);
                break;
            // Section 2: Incremental Marks Entry ("Then and Now")
            case "9":
                academicDAO.enterPhase1CIA(sc);
                break;
            case "10":
                academicDAO.enterPhase2Activities(sc);
                break;
            case "11":
                academicDAO.enterPhase3ESE(sc);
                break;
            // Section 3: Academic Reports (KEC R2024)
            case "12":
                academicDAO.generateStudentMarkSheet(sc);
                break;
            case "13":
                academicDAO.generateAttendanceDefaulterReport();
                break;
            case "14":
                academicDAO.generateClassPerformanceSummary();
                break;
            case "0":
                System.out.println("\nExiting KEC Academic & Continuous Evaluation System.");
                System.out.println("Kongu Engineering College - Transform Yourself. Goodbye!\n");
                running = false;
                break;
            default:
                System.out.println("\n[Invalid Selection] Please choose a number from the menu (0 to 14).");
                break;
        }
        if (running) {
            System.out.print("\nPress [Enter] to return to Main Menu...");
            sc.nextLine();
        }
    }
    sc.close();
}
private static void printBanner() {
    System.out.println("============================================================
        ======================");
    System.out.println("       KONGU ENGINEERING COLLEGE (AUTONOMOUS) - PERUNDURAI,
        ERODE");
    System.out.println("          REGULATION 2024 (R2024) - STUDENT ACADEMIC EVALUATION
        SYSTEM");
    System.out.println("                    JAVA JDBC IMPLEMENTATION");
```

```
        System.out.println("============================================================
            ======================");
    }
    private static void printMenu() {
        System.out.println("\n============================================================
            ========================");
        System.out.println("                          MAIN MENU");
        System.out.println("============================================================
            ======================");
        System.out.println(" [STUDENT & COURSE CATALOG MANAGEMENT]");
        System.out.println("   1. Add New Student Registration");
        System.out.println("   2. View Registered Students Directory");
        System.out.println("   3. Add Course to Catalog (THEORY / PRACTICAL / EMBEDDED)");
        System.out.println("   4. View Course Catalog (Credits & Weightage Rules)");
        System.out.println("   5. Enroll Student in Course");
        System.out.println("   6. View Course Enrollments & Attendance Status");
        System.out.println("   7. Continuously Record / Update Attendance (Held vs Attended)");
        System.out.println("   8. Approve Medical Condonation (65% to 74.9% Eligibility)");
        System.out.println("-------------------------------------------------------------------------------");
        System.out.println(" [INCREMENTAL MARKS ENTRY (THEN AND NOW)]");
        System.out.println("   9. Phase 1 (Mid-Semester)       : Enter CIA Test 1 & CIA Test 2 (0-100)");
        System.out.println("  10. Phase 2 (End of Semester - CIA): Enter Assignments / Quizzes / Lab (0-100)");
        System.out.println("  11. Phase 3 (Post-Semester)      : Enter ESE Marks (0-100) or Mark Absent (AB)");
        System.out.println("-------------------------------------------------------------------------------");
        System.out.println(" [FORMAL ACADEMIC REPORTING]");
        System.out.println("  12. Student Semester Mark Sheet (Detailed Grades, SGPA, & Percentage)");
        System.out.println("  13. Attendance Defaulter Report (Attendance < 65% Barred & Awarded SA)");
        System.out.println("  14. Class Performance & Result Summary (Dept Pass/Fail Count & Avg SGPA)");
        System.out.println("-------------------------------------------------------------------------------");
        System.out.println("   0. Exit Application");
        System.out.println("============================================================
            ======================");
    }
}
```

DB CONNECTION.JAVA:

```
import java.io.File;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;
/**
 * DBConnection provides JDBC database connectivity for the KEC Academic System.
 * Supports automatic self-healing: if the local database engine on port 3307 is stopped,
 * DBConnection automatically launches the background mysqld process and connects.
 */
public final class DBConnection {
    private static final String DEFAULT_HOST = "localhost";
    private static final String DEFAULT_DB = "kec_academic";
    private static final String DEFAULT_USER = "root";
```

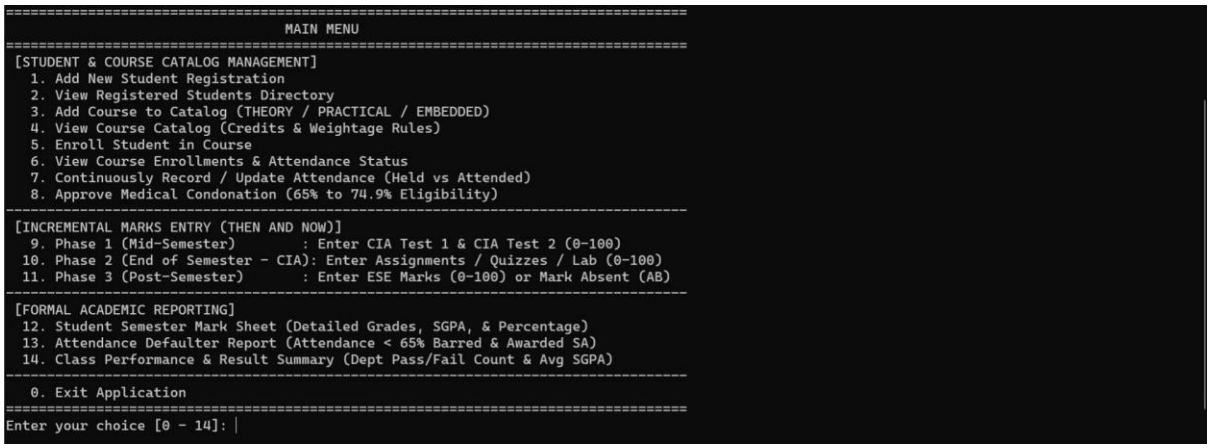
```
private static final String DEFAULT_PASSWORD = "";
private static volatile boolean autoStartAttempted = false;
// Private constructor to prevent instantiation
private DBConnection() {}
/**
 * Establishes and returns a database connection.
 * Tries port 3307 first (with auto-start self-healing), then falls back to 3306.
 *
 * @return active java.sql.Connection
 * @throws SQLException if connection cannot be established
 */
public static Connection getConnection() throws SQLException {
    String explicitUrl = System.getenv("KEC_DB_URL");
    String user = getEnv("KEC_DB_USER", DEFAULT_USER);
    String password = getEnv("KEC_DB_PASSWORD", DEFAULT_PASSWORD);
    if (explicitUrl != null && !explicitUrl.trim().isEmpty()) {
        return DriverManager.getConnection(explicitUrl.trim(), user, password);
    }
    String host = getEnv("KEC_DB_HOST", DEFAULT_HOST);
    String db = getEnv("KEC_DB_NAME", DEFAULT_DB);
    String portEnv = System.getenv("KEC_DB_PORT");
    if (portEnv != null && !portEnv.trim().isEmpty()) {
        String url = buildUrl(host, portEnv.trim(), db);
        return DriverManager.getConnection(url, user, password);
    }
    // Try port 3307 first
    String url3307 = buildUrl(host, "3307", db);
    try {
        return DriverManager.getConnection(url3307, user, password);
    } catch (SQLException ex3307) {
        // Port 3307 is not responding. Try to auto-start local mysqld if not yet attempted.
        if (!autoStartAttempted) {
            autoStartAttempted = true;
            if (startLocalDatabase()) {
                // Try connecting to 3307 again after auto-starting
                for (int retry = 0; retry < 5; retry++) {
                    try {
                        Thread.sleep(1000);
                        return DriverManager.getConnection(url3307, user, password);
                    } catch (Exception ignored) {
                    }
                }
            }
        }
    }
    // Fallback to port 3306 if 3307 still cannot be contacted
    String url3306 = buildUrl(host, "3306", db);
    try {
        return DriverManager.getConnection(url3306, user, password);
    } catch (SQLException ex3306) {
```

```
            throw new SQLException("Failed to connect to MySQL database '" + db + "' on ports 3307 and
                3306.\n"
                    + "Ensure MySQL is running or configure KEC_DB_URL / KEC_DB_PASSWORD in the
                        environment.\n"
                    + "Details: " + ex3306.getMessage(), ex3306);
        }
    }
    private static boolean startLocalDatabase() {
        try {
            String exe = "C:\\Program Files\\MySQL\\MySQL Server 26.10\\bin\\mysqld.exe";
            File exeFile = new File(exe);
            if (!exeFile.exists()) return false;
            File dataDir = new File(System.getProperty("user.dir"), "test_db");
            if (!dataDir.exists()) {
                dataDir = new File("C:\\Users\\suriy\\Downloads\\KECEval\\test_db");
            }
            if (!dataDir.exists()) return false;
            ProcessBuilder pb = new ProcessBuilder(
                    exe,
                    "--datadir=" + dataDir.getAbsolutePath(),
                    "--port=3307",
                    "--mysqlx-port=33070"
            );
            pb.redirectErrorStream(true);
            pb.start();
            return true;
        } catch (Exception e) {
            return false;
        }
    }
    private static String buildUrl(String host, String port, String db) {
        return "jdbc:mysql://" + host + ":" + port + "/" + db
                + "?useSSL=false&allowPublicKeyRetrieval=true&serverTimezone=UTC";
    }
    private static String getEnv(String key, String fallback) {
        String val = System.getenv(key);
        return (val != null) ? val : fallback;
    }
    /**
     * Utility method to test database connectivity.
     *
     * @return true if connected successfully, false otherwise.
     */
    public static boolean testConnection() {
        try (Connection con = getConnection()) {
            return con != null && !con.isClosed();
        } catch (SQLException e) {
            System.err.println("[DBConnection] Connection test failed: " + e.getMessage());
            return false;
        }
```

```
    }
}
```

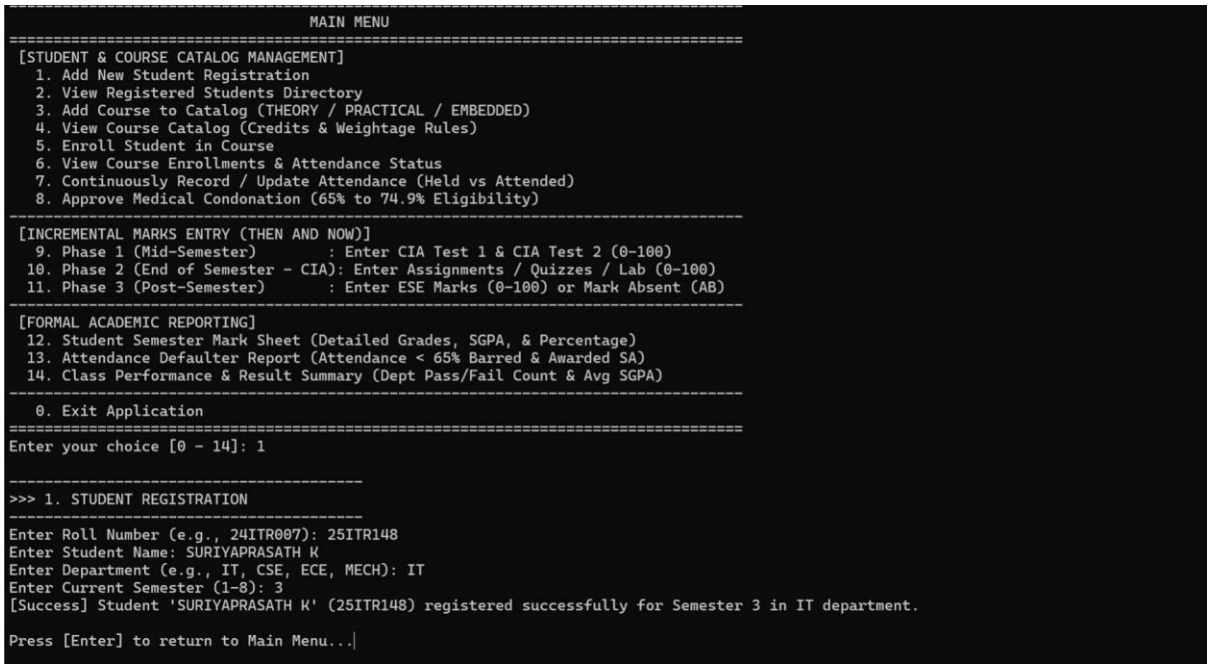
OUTPUT SCREENSHOTS:

MAIN MENU:

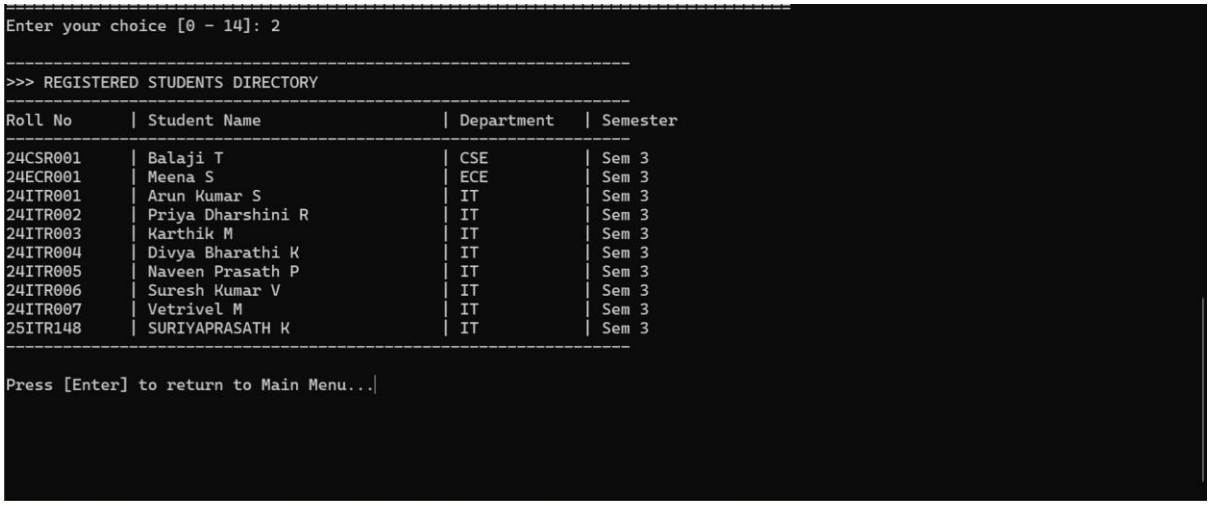


STUDENT & COURSE MANAGEMENT:

1. STUDENT REGISTRATION:



2. REGISTERED STUDENTS DIRECTORY:

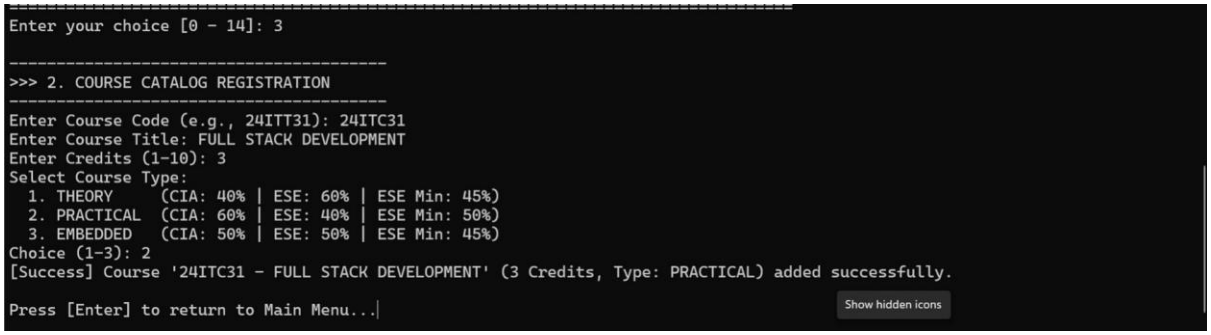


```
Enter your choice [0 - 14]: 2

------------------------------------------------------------
>>> REGISTERED STUDENTS DIRECTORY
------------------------------------------------------------
Roll No      | Student Name                | Department   | Semester
------------------------------------------------------------
24CSR001     | Balaji T                    | CSE          | Sem 3
24ECR001     | Meena S                     | ECE          | Sem 3
24ITR001     | Arun Kumar S                | IT           | Sem 3
24ITR002     | Priya Dharshini R           | IT           | Sem 3
24ITR003     | Karthik M                   | IT           | Sem 3
24ITR004     | Divya Bharathi K            | IT           | Sem 3
24ITR005     | Naveen Prasath P            | IT           | Sem 3
24ITR006     | Suresh Kumar V              | IT           | Sem 3
24ITR007     | Vetrivel M                  | IT           | Sem 3
25ITR148     | SURIYAPRASATH K             | IT           | Sem 3
------------------------------------------------------------

Press [Enter] to return to Main Menu...
```

3. ADD COURSE TO CATALOG:

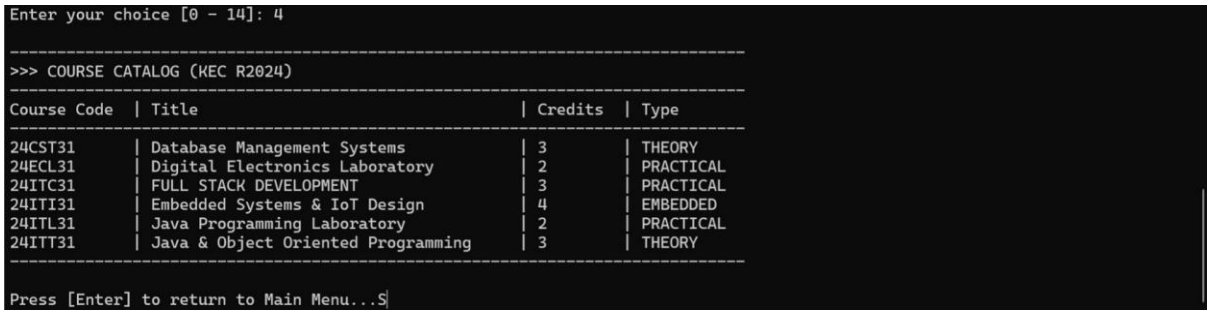


```
Enter your choice [0 - 14]: 3

----------------------------------------
>>> 2. COURSE CATALOG REGISTRATION
----------------------------------------
Enter Course Code (e.g., 24ITT31): 24ITC31
Enter Course Title: FULL STACK DEVELOPMENT
Enter Credits (1-10): 3
Select Course Type:
  1. THEORY     (CIA: 40% | ESE: 60% | ESE Min: 45%)
  2. PRACTICAL  (CIA: 60% | ESE: 40% | ESE Min: 50%)
  3. EMBEDDED   (CIA: 50% | ESE: 50% | ESE Min: 45%)
Choice (1-3): 2
[Success] Course '24ITC31 - FULL STACK DEVELOPMENT' (3 Credits, Type: PRACTICAL) added successfully.

Press [Enter] to return to Main Menu...
```

4. VIEW COURSE CATALOG:

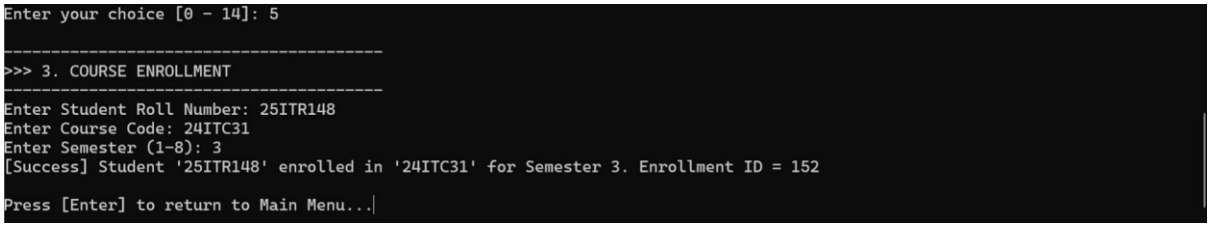


```
Enter your choice [0 - 14]: 4

------------------------------------------------------------------------------
>>> COURSE CATALOG (KEC R2024)
------------------------------------------------------------------------------
Course Code  | Title                                | Credits  | Type
------------------------------------------------------------------------------
24CST31      | Database Management Systems          | 3        | THEORY
24ECL31      | Digital Electronics Laboratory       | 2        | PRACTICAL
24ITC31      | FULL STACK DEVELOPMENT               | 3        | PRACTICAL
24ITI31      | Embedded Systems & IoT Design        | 4        | EMBEDDED
24ITL31      | Java Programming Laboratory          | 2        | PRACTICAL
24ITT31      | Java & Object Oriented Programming   | 3        | THEORY
------------------------------------------------------------------------------

Press [Enter] to return to Main Menu...S
```

5. COURSE ENROLLMENT:



```
Enter your choice [0 - 14]: 5

----------------------------------------
>>> 3. COURSE ENROLLMENT
----------------------------------------
Enter Student Roll Number: 25ITR148
Enter Course Code: 24ITC31
Enter Semester (1-8): 3
[Success] Student '25ITR148' enrolled in '24ITC31' for Semester 3. Enrollment ID = 152

Press [Enter] to return to Main Menu...
```

6. VIEW ENROLLMENTS & ATTENDANCE STATUS:

```
Enter your choice [0 - 14]: 6

-----------------------------------------------------------------------------------------------
>>> COURSE ENROLLMENTS & ATTENDANCE STATUS
-----------------------------------------------------------------------------------------------
EID   | Roll No    | Student Name         | Course   | Sem    | Attended  | Percent  | Eligibility
-----------------------------------------------------------------------------------------------
101   | 24ITR001   | Arun Kumar S         | 24ITT31  | Sem 3  |  48/50    |  96.00% | Eligible
102   | 24ITR001   | Arun Kumar S         | 24ITL31  | Sem 3  |  29/30    |  96.67% | Eligible
103   | 24ITR001   | Arun Kumar S         | 24ITI31  | Sem 3  |  57/60    |  95.00% | Eligible
104   | 24ITR002   | Priya Dharshini R    | 24ITT31  | Sem 3  |  42/50    |  84.00% | Eligible
105   | 24ITR002   | Priya Dharshini R    | 24ITL31  | Sem 3  |  26/30    |  86.67% | Eligible
106   | 24ITR002   | Priya Dharshini R    | 24ITI31  | Sem 3  |  49/60    |  81.67% | Eligible
107   | 24ITR003   | Karthik M            | 24ITT31  | Sem 3  |  28/50    |  56.00% | SA (Barred <65%)
108   | 24ITR003   | Karthik M            | 24ITL31  | Sem 3  |  24/30    |  80.00% | Eligible
109   | 24ITR004   | Divya Bharathi K     | 24ITT31  | Sem 3  |  35/50    |  70.00% | Medical Condoned
110   | 24ITR004   | Divya Bharathi K     | 24ITL31  | Sem 3  |  27/30    |  90.00% | Eligible
111   | 24ITR005   | Naveen Prasath P     | 24ITT31  | Sem 3  |  45/50    |  90.00% | Eligible
112   | 24ITR006   | Suresh Kumar V       | 24ITT31  | Sem 3  |  40/50    |  80.00% | Eligible
113   | 24CSR001   | Balaji T             | 24CST31  | Sem 3  |  45/50    |  90.00% | Eligible
114   | 24ECR001   | Meena S              | 24ECL31  | Sem 3  |  28/30    |  93.33% | Eligible
115   | 24ITR007   | Vetrivel M           | 24ITT31  | Sem 3  |  45/50    |  90.00% | Eligible
152   | 25ITR148   | SURIYAPRASATH K      | 24ITC31  | Sem 3  |   0/0     |   0.00% | Not Recorded
-----------------------------------------------------------------------------------------------

Press [Enter] to return to Main Menu...
```

7. ATTENDANCE TRACKING:

```
Enter your choice [0 - 14]: 7

 ----------------------------------------
 >>> 4. ATTENDANCE TRACKING
 ----------------------------------------
 Enter Enrollment ID or Roll Number (e.g., 25ITR148): 25ITR148

 Found 4 enrolled course(s) for SURIYAPRASATH K (25ITR148):
   1. 24ITC31  - FULL STACK DEVELOPMENT           (PRACTICAL, Sem 3) [EID: 152]
   2. 24ITT31  - Java & Object Oriented Programming (THEORY   , Sem 3) [EID: 153]
   3. 24ITL31  - Java Programming Laboratory      (PRACTICAL, Sem 3) [EID: 154]
   4. 24ITI31  - Embedded Systems & IoT Design    (EMBEDDED , Sem 3) [EID: 155]
 Select course (1-4, or enter Enrollment ID, or 0 to cancel): 2
 >> Selected: 24ITT31 - Java & Object Oriented Programming [Enrollment ID: 153]
 Enter Total Classes Held: 7
 Enter Total Classes Attended: 4
 [Success] Attendance updated for Enrollment ID 153: 4/7 (57.14%)
 >> Status: SHORTAGE OF ATTENDANCE (SA). Barred from ESE (Awarded SA Grade).

 Press [Enter] to return to Main Menu...
```

ATTENDANCE & GRADING RULES:

1. MEDICAL CONDONATION:

```
Enter your choice [0 - 14]: 8

----------------------------------------
>>> 8. APPROVE MEDICAL CONDONATION
----------------------------------------
Enter Enrollment ID or Roll Number (e.g., 25ITR148) to approve: 25ITR148

Found 4 enrolled course(s) for SURIYAPRASATH K (25ITR148):
  1. 24ITC31  - FULL STACK DEVELOPMENT           (PRACTICAL, Sem 3) [EID: 152]
  2. 24ITT31  - Java & Object Oriented Programming (THEORY   , Sem 3) [EID: 153]
  3. 24ITL31  - Java Programming Laboratory      (PRACTICAL, Sem 3) [EID: 154]
  4. 24ITI31  - Embedded Systems & IoT Design    (EMBEDDED , Sem 3) [EID: 155]
Select course (1-4, or enter Enrollment ID, or 0 to cancel): 2
>> Selected: 24ITT31 - Java & Object Oriented Programming [Enrollment ID: 153]
Student: SURIYAPRASATH K (25ITR148) | Course: 24ITT31 | Attendance: 48/50 (96.00%)
[Notice] Attendance is 96.00% (>=75%). Student is already fully eligible; condonation is not required.
```

INCREMENTAL MARKS ENTRY:

1. PHASE 1 (MID-SEMESTER):

```
Enter your choice [0 - 14]: 9

------------------------------------------------------
>>> 9. PHASE 1: MID-SEMESTER MARKS (CIA 1 & CIA 2)
------------------------------------------------------
Enter Enrollment ID or Roll Number (e.g., 25ITR148): 25ITR148

Found 4 enrolled course(s) for SURIYAPRASATH K (25ITR148):
  1. 24ITC31  - FULL STACK DEVELOPMENT           (PRACTICAL, Sem 3) [EID: 152]
  2. 24ITT31  - Java & Object Oriented Programming (THEORY   , Sem 3) [EID: 153]
  3. 24ITL31  - Java Programming Laboratory      (PRACTICAL, Sem 3) [EID: 154]
  4. 24ITI31  - Embedded Systems & IoT Design    (EMBEDDED , Sem 3) [EID: 155]
Select course (1-4, or enter Enrollment ID, or 0 to cancel): 2
>> Selected: 24ITT31 - Java & Object Oriented Programming [Enrollment ID: 153]
Student: SURIYAPRASATH K (25ITR148) | Course: 24ITT31 - Java & Object Oriented Programming (THEORY)
Enter CIA Test 1 Mark (0-100): 70
Enter CIA Test 2 Mark (0-100): 65
[Success] Phase 1 CIA marks recorded for Enrollment ID 153 (CIA1: 70.00, CIA2: 65.00)
```

2. PHASE 2 (END OF SEMESTER - CIA):

```
Enter your choice [0 - 14]: 10

------------------------------------------------------------
>>> 10. PHASE 2: END-OF-SEMESTER CIA (ASSIGNMENTS/ACTIVITIES)
------------------------------------------------------------
Enter Enrollment ID or Roll Number (e.g., 25ITR148): 25ITR148

Found 4 enrolled course(s) for SURIYAPRASATH K (25ITR148):
  1. 24ITC31  - FULL STACK DEVELOPMENT           (PRACTICAL, Sem 3) [EID: 152]
  2. 24ITT31  - Java & Object Oriented Programming (THEORY   , Sem 3) [EID: 153]
  3. 24ITL31  - Java Programming Laboratory      (PRACTICAL, Sem 3) [EID: 154]
  4. 24ITI31  - Embedded Systems & IoT Design    (EMBEDDED , Sem 3) [EID: 155]
Select course (1-4, or enter Enrollment ID, or 0 to cancel): 2
>> Selected: 24ITT31 - Java & Object Oriented Programming [Enrollment ID: 153]
Student: SURIYAPRASATH K (25ITR148) | Course: 24ITT31 - Java & Object Oriented Programming (THEORY)
Enter Assignments / Quizzes / Lab Evaluations Mark (0-100): 88
[Success] Phase 2 activities mark recorded for Enrollment ID 153 (Activities: 88.00)
```

3. PHASE 3 (POST SEMESTER - ESE / ABSENT):

```
Enter your choice [0 - 14]: 11

------------------------------------------------------------
>>> 11. PHASE 3: POST-SEMESTER END SEMESTER EXAM (ESE / AB)
------------------------------------------------------------
Enter Enrollment ID or Roll Number (e.g., 25ITR148): 25ITR148

Found 4 enrolled course(s) for SURIYAPRASATH K (25ITR148):
  1. 24ITC31  - FULL STACK DEVELOPMENT           (PRACTICAL, Sem 3) [EID: 152]
  2. 24ITT31  - Java & Object Oriented Programming (THEORY   , Sem 3) [EID: 153]
  3. 24ITL31  - Java Programming Laboratory      (PRACTICAL, Sem 3) [EID: 154]
  4. 24ITI31  - Embedded Systems & IoT Design    (EMBEDDED , Sem 3) [EID: 155]
Select course (1-4, or enter Enrollment ID, or 0 to cancel): 2
>> Selected: 24ITT31 - Java & Object Oriented Programming [Enrollment ID: 153]
Student: SURIYAPRASATH K (25ITR148) | Course: 24ITT31 - Java & Object Oriented Programming (THEORY)
Was the student ABSENT in ESE? (Y/N): Y
[Success] Enrollment ID 153 marked as ABSENT (AB). Grade will be AB (0 Grade Points).
```

FORMAL ACADEMIC REPORTING:

STUDENT SEMESTER MARK SHEET:

```
Enter your choice [0 - 14]: 12

------------------------------------------------------------
>>> 12. STUDENT SEMESTER ACADEMIC MARK SHEET (KEC R2024)
------------------------------------------------------------
Enter Student Roll Number: 25ITR148

=============================================================================================================
                          KONGU ENGINEERING COLLEGE (AUTONOMOUS) - PERUNDURAI
                              REGULATION 2024 (R2024) - GRADE REPORT
=============================================================================================================
Roll Number : 25ITR148          Student Name : SURIYAPRASATH K                Department : IT
Semester    : 3                 Academic Standing : Autonomous Affiliated       Regulation : R2024
-------------------------------------------------------------------------------------------------------------
Code      | Course Title               | Type      | Crd | Attd%  | CIA/W  | ESE/W  | Tot/100 | Grade | GP  | Status
-------------------------------------------------------------------------------------------------------------
24ITC31   | FULL STACK DEVELOPMENT     | PRACTICAL | 3   |   0.0% | --     | --     | --      | PENDING | 0   | PENDING
24ITI31   | Embedded Systems & IoT De. | EMBEDDED | 4   |   0.0% | --     | --     | --      | PENDING | 0   | PENDING
24ITL31   | Java Programming Laborato. | PRACTICAL | 2   |  70.0% | 54.0   | 34.0   | 88.00   | A+    | 9   | PASS
24ITT31   | Java & Object Oriented Pr. | THEORY    | 3   |  96.0% | --     | AB     | --      | AB    | 0   | AB
-------------------------------------------------------------------------------------------------------------
Credits Enrolled: 5  |  Credits Earned: 2  |  Total Grade Points Earned: 18.0
-------------------------------------------------------------------------------------------------------------
SEMESTER 3 SGPA : 3.60 / 10.00         EQUIVALENT PERCENTAGE : 36.00%
Academic Status : RE-APPEARANCE / CLEARANCE REQUIRED
[Note] One or more course assessments or attendance approvals are pending finalization.
=============================================================================================================
```

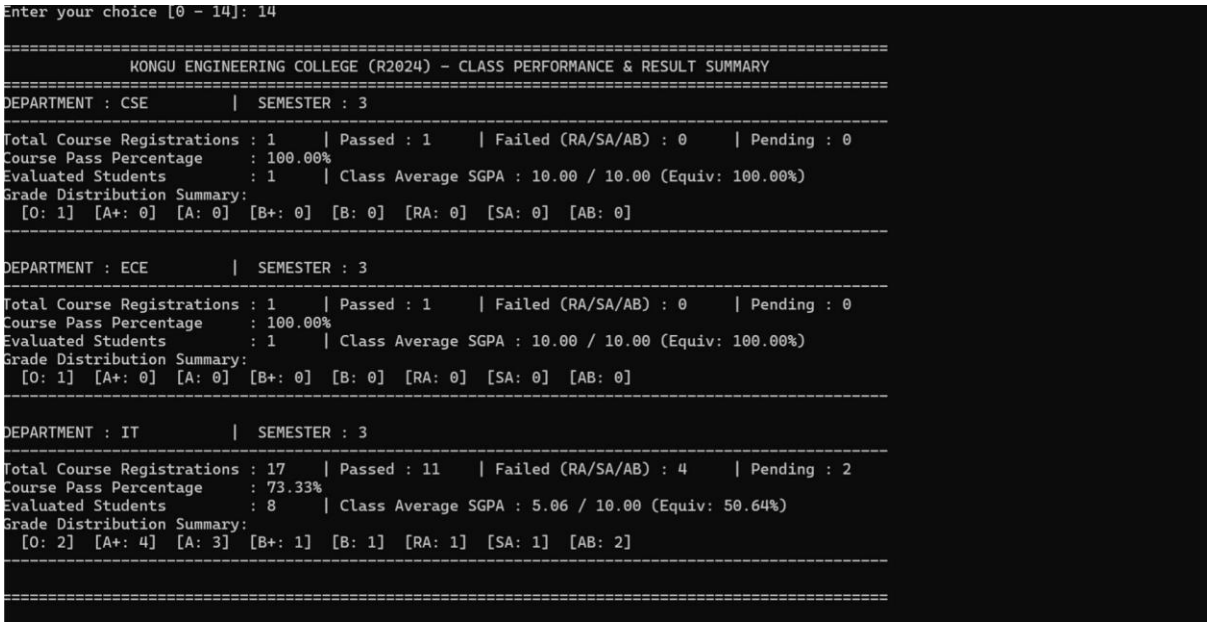
ATTENDANCE DEFAULTER REPORT:

```
Enter your choice [0 - 14]: 13

=============================================================================================
            KONGU ENGINEERING COLLEGE (R2024) - ATTENDANCE DEFAULTER REPORT
                 [Students with Attendance < 65% Barred from ESE (Awarded SA)]
=============================================================================================
Roll No    | Student Name           | Dept   | Course    | Course Title             | Attendance | Grade  | Status
---------------------------------------------------------------------------------------------
24ITR003   | Karthik M              | IT     | 24ITT31   | Java & Object Oriented . | 28/50 (56.0%) | SA     | BARRED
---------------------------------------------------------------------------------------------
Total Attendance Defaulters Awarded SA Grade: 1

---------------------------------------------------------------------------------------------
>>> CONDITIONAL ATTENDANCE TRACKER (65% to 74.9% - Medical Condonation Review)
---------------------------------------------------------------------------------------------
Roll: 24ITR004   | Name: Divya Bharathi K     | Course: 24ITT31  | Attendance: 35/50 (70.00%) | Medical Approved: YES | Status: ELIGI
BLE VIA CONDONATION
Roll: 25ITR148   | Name: SURIYAPRASATH K      | Course: 24ITL31  | Attendance: 35/50 (70.00%) | Medical Approved: YES | Status: ELIGI
BLE VIA CONDONATION
=============================================================================================
```

DEPARTMENT-WISE CLASS PERFORMANCE & RESULT SUMMARY:



RESULT:

Thus, the Student Academic and Continuous Evaluation System for Kongu Engineering College (Autonomous) Regulation 2024 was successfully designed, developed, and tested using Java (JDBC) and MySQL. The system automates the calculation of Continuous Internal Assessments (CIA), End Semester Examinations (ESE), letter grades, attendance eligibility, and Semester Grade Point Average (SGPA).

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5. University Grants Commission, Guidelines for Choice Based Credit System (CBCS).
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9. YouTube, Java JDBC with MySQL Tutorial Videos.
https://www.youtube.com/results?search_query=Java+JDBC+MySQL+tutorial .

10. Anthropic, Claude AI Assistant (used for project ideas and guidance).
https://claude.ai/ .

GITHUB:

LINK:	https://github.com/suriyaprasath8c-dotcom/Java-Individual-Project